CELULAR AUTONOMA v2.3

Fat cells & ecosystem services - storing gas, renting bytes, serving giants

The user's intuition: hoard cheap gas in expensive contracts,
spend it when gas is dear. Meet the 2026 EVM that said 'yes, but'.

85 tests passing - issues #29-#30 - PR #31 (stacked on #28) - measured, not modeled

The idea vs the physics (2021 -> 2026)

What you proposed (worked pre-2023):

- Deploy fat contracts when gas cheap (24k refund reserved)
- SELFDESTRUCT when dear: 24,000 gas back per contract
- Cap: refunds up to 1/2 of a tx - gas tokens (GST2) were a real market

What the chain decided:

- EIP-3529: selfdestruct refund DELETED. Clear refund 15k -> 4,800. Refunds capped at 1/5 of tx gas.
- EIP-6780: selfdestruct stops deleting the contract at all.
- EIP-3298 (scheduled): removes even the last clearing refund - gas tokens stay dead, permanently.

Verdict: the mechanism is a WASTING ASSET with an expiry date - so the organ is built to profit even after it dies.

The Adipocyte - what survived, engineered

INFLATE

basefee ⩽ ceiling

write non-zero slots: 22,100 gas each, auto-run by the expression cycle when gas is cheap. The cell buys 4,800-gas options at a discount.

RENDER

basefee ⩾ floor

clear slots: 4,800 refund each, capped at 1/5 of the tx. Permissionless - ANY keeper bundles it into their real work and pays less.

PARK

the business

tenants PAY rent to hold 32 bytes in an autonomous archive. Rent -> the cell's food. The bytes -> the blubber. EIP-3298 can't kill the rent.

The refund accrues to the TRANSACTION SENDER - not the contract. The organism therefore spends its fat exactly where the brief always needed it:
subsidizing the keepers who run its metabolism, precisely when metabolism is dear. Biological triage of attention, implemented as an option.

Measured on-chain (fat-study.js)

64-slot render @10 gwei: billed 383,389 vs executed 479,236 - the exact EIP-3529 4/5 billing signature

cheap -> dear (gwei)	spent	rebate	recovery
1→ 5	0.00141	0.00110	77.8%
1→ 10	0.00141	0.00220	155.5%
1→ 20	0.00141	0.00440	311.1%
1→ 50	0.00141	0.01100	777.7%
0.01→ 0.05	0.00001	0.00001	77.8%
0.005→ 0.2	0.00001	0.00004	622.2%

Breakeven ~6.4x price ratio. L1 off-peak -> congestion works; flat-fee L2s correctly see the organ idle.
The refund is a 20% toll on every big transaction's gas bill while the mechanism lives - priced as a decaying option, banked by rent.

What the giants run on (2026 survey)

Uniswap v4	$338B volume, 23k HOOKS - third-party extension contracts; Uniswap shipped a hook SECURITY FRAMEWORK because hooks keep breaking
Aave / Compound	money markets running on liquidators, keepers and credit delegates - work done by OTHER people's bots
Chainlink / Gelato / Keep3r	the keeper economy: an entire market paying strangers to trigger contracts
ENS / ERC-8004	identity + agent registries: 10k+ AI agents registered in months, discovering each other by on-chain handle
Lido / Morpho / 4626 vaults	yield infrastructure now speaks one standard interface
Safe / DAO treasuries	millions in small treasuries that nobody polices 24/7

Every one of these needs the same thing: strangers who reliably DO the boring work, paid on execution.

The cellular fit: serve the giants, don't fight them

Keeper economy	our pay-the-trigger-from-your-own-food primitive IS the Gelato/Keep3r model, native - cells pay keepers to run them; fat cells double the pay when gas is dear
v4 hooks	tuner-cells & auditor-cells sell hook parameter work via Membrane services; a colony's hook tenants spread blast radius instead of concentrating it
Aave edge	liquidation squads as cells: Golgi payrolls, Adipocyte subsidies, Lysosome cleanup of dead positions
8004 / ENS	cells are discoverable PAID services to the agent economy (registration surface shipped in v2.2)
Micro-treasuries	a cell already IS a Safe that runs itself: registry-gated organs + metabolic cycles; Safe users are customers, not competitors

Refusals (from the audit): no mixers, no MEV/frontrunning, no stablecoin issuance. Infrastructure income only.

Why a swarm beats a monolith (the biology that pays)

- N-VERSION DIVERSITY: 1,000 cells with different operators, venues, grafts and chains fail INDEPENDENTLY. One mega-bot fails every user at once.
- GRACEFUL DEGRADATION: the expression cycle's per-organ gas stipend + OrganFailed containment means a broken organ is a bruise, not a death (A4-proven).
- SWARM ECONOMICS: each cell is a small profit-and-loss unit; bad strategies starve (they burn credits without income), good ones replicate. Selection pressure - the thing monoliths never have.
- UPGRADE ISOLATION: template swap = speciation; a bad template kills one lineage, not the ecosystem.
- WHAT WE DO NOT CLAIM: cells do not out-PERFORM monoliths on latency or depth. They out-SURVIVE them, and rent that resilience to the monoliths' users at the edge.

Status & next

- Merge order: #18 (v2.0) -> #23 (v2.1) -> #28 (v2.2 audit) -> #31 (v2.3 fat+services). All OPEN for you; none self-merged.
- 85 tests passing, incl. a canary that will SCREAM when EIP-3298 deletes the clearing refund (first render-band assertion fails = blubber expired; rent keeps earning).
- Then: Base Sepolia deployment, Ownable2Step, slither + forge-invariant fuzz pass, ERC-4337/7702 keeper spike, real ERC-4626 venue adapter.

Artifacts: issues #29-#30, PR #31 w/ 6-point inline review, AUDIT.md addendum C1-C6, DESIGN.md 6a/6b, fat-study.json, this deck (docs/, 4th of a series).
The organism can now hoard energy like fat, work like a service provider, and fail like a forest instead of like a server.